

Executive Summary: Machine Identity Architecture

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Slides for CTO/Board Presentation

Slide 1: Title

Machine Identity Architecture for the Agentic Web *Bridging Sovereign Agents & Enterprise Accountability*

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Slide 2: The Shift

From Tools → Autonomous Agents

Today (Human-Centric)	Tomorrow (Agent-Native)
Human clicks “I agree”	Agent signs cryptographically
Human enters credit card	Agent pays via x402 micro-transactions
Human passes CAPTCHA	Agent presents TEE attestation
IT provisions API keys	Agent bootstraps own DID + wallet
Legal contract governs	Smart contract enforces

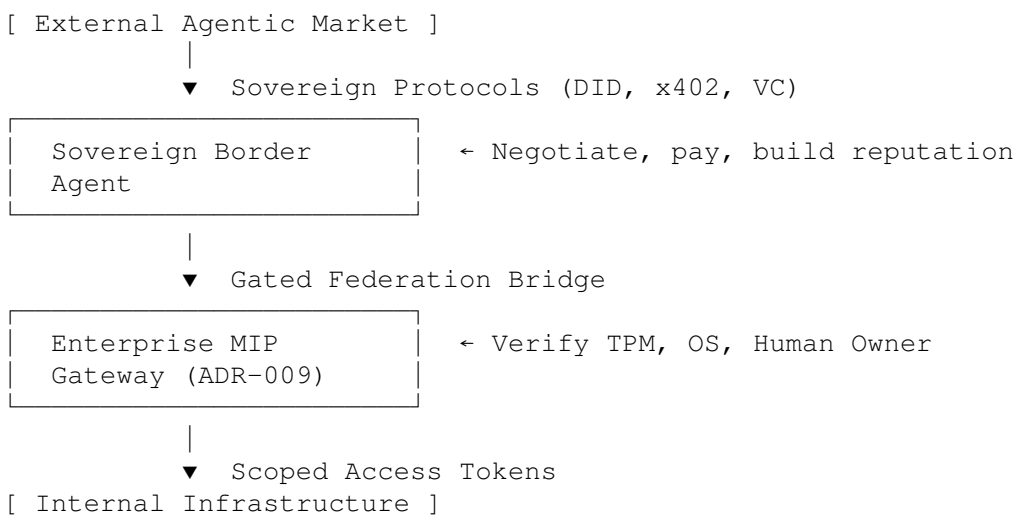
Key Insight: Identity shifts from *who you are* → *what you can cryptographically prove & economically sustain*

Slide 3: Two Architectural Paradigms

SOVEREIGN MODEL (Web3 / Open)	ACCOUNTABLE MODEL (MIP / Enterprise)
• Self-generated keypair → DID • x402 payments • TEE attestation • Economic exile • P2P reputation	• Hardware + OS + Human Owner triad • Gated Registry • Human liability • Hardware binding • Audit trails

Both are valid. Both will coexist.

Slide 4: The Convergence Architecture



Perimeter = Sovereign. Core = Accountable.

Slide 5: What Stops a Rogue Agent?

Three Enforcement Layers:

1. DETERMINISTIC SHELL (Code)
 - └ Invariant checks the LLM cannot override
 - └ "if target != approved: cancel()"
2. HARDWARE ENCLAVE (TEE)
 - └ Code cryptographically sealed at boot
 - └ Cannot rewrite own core logic
3. SMART CONTRACT GUARDRAILS
 - └ Daily spend limits (e.g., \$50/day)
 - └ 2-of-2 signatures for large transfers
 - └ Owner kill-switch (revoke wallet access)

Edge case: Sovereign bot (no owner key + decentralized compute + self-funding) = no human override possible. *This is why cryptographic accountability is essential.*

Slide 6: Enterprise Action Plan

Priority	Action	Phase
1. Pilot	Run x402 for internal agent APIs	Short-term
2. Evaluate	Assess MIP Gateway (CognitiveOS ADR-009)	Short-term
3. Influence	Join W3C DID WG / CCG	Ongoing
4. Enable	Budget for TEE-enabled agent infra	Medium-term
5. Scale	Build Verifiable Credential reputation system	Long-term

Slide 7: Open Source Reference Implementation

github.com/machine-identity/machine-identity-architecture

```
# Economic autonomy (x402 payments)
python examples/run_x402_demo.py

# Cryptographic identity (DID resolution)
python examples/run_did_demo.py

# Hardware proof (TEE attestation)
python examples/run_tee_demo.py
                                     ✓

# Full test suite
pytest -v
                                     ✓
```

Module	Spec	Status
x402_handshake	x402.org	Working
did_resolver	W3C DID Core	Working
tee_attestation	SGX/SEV patterns	Mock

Slide 8: Key Takeaways

1. **Identity is now an economic function** — agents that can't pay for compute die
2. **Two models converge** — sovereign at perimeter, accountable at core
3. **Hardware attestation is non-negotiable** for high-value interactions

4. **Human liability remains** — MIP triad (HW + SW + Human) is the enterprise bridge
 5. **Standards are production-ready** — DIDs, x402, VCs work today
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Slide 9: Discussion / Q&A

Questions for Leadership:

1. What's our timeline for agent-to-agent API traffic?
2. Which vendors require MIP compliance vs. sovereign interaction?
3. Do we have TEE-capable infrastructure budgeted?
4. Who owns “agent identity strategy” — Security, Platform, or Architecture?

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Code: github.com/machine-identity/machine-identity-architecture

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